

```
C:\Users\Jane Eyre\Desktop\Python-Study>python
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Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
```

```
>>> states = ['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California', 'Colorado',
...           'Connecticut', 'Delaware', 'Florida', 'Georgia', 'Hawaii', 'Idaho',
...           'Illinois', 'Indiana', 'Iowa', 'Kansas', 'Kentucky', 'Louisiana',
...           'Maine', 'Maryland', 'Massachusetts', 'Michigan', 'Minnesota',
...           'Mississippi', 'Missouri', 'Montana', 'Nebraska', 'Nevada',
...           'New Hampshire', 'New Jersey', 'New Mexico', 'New York',
...           'North Carolina', 'North Dakota', 'Ohio', 'Oklahoma', 'Oregon',
...           'Pennsylvania', 'Rhode Island', 'South Carolina', 'South Dakota',
...           'Tennessee', 'Texas', 'Utah', 'Vermont', 'Virginia', 'Washington',
...           'West Virginia', 'Wisconsin', 'Wyoming']
>>> states[0]
'Alabama'
>>> states[-1]
'Wyoming'
>>> for s in states:
...     if ' ' in s:
...         print(s)
...
New Hampshire
New Jersey
New Mexico
New York
North Carolina
North Dakota
Rhode Island
South Carolina
South Dakota
West Virginia
>>> [s for s in states if ' ' in s]
['New Hampshire', 'New Jersey', 'New Mexico', 'New York', 'North Carolina', 'North Dakota', 'Rhode Island', 'South Carolina', 'South Dakota', 'West Virginia']
>>> [s for s in states]
['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California', 'Colorado', 'Connecticut', 'Delaware', 'Florida', 'Georgia', 'Hawaii', 'Idaho', 'Illinois', 'Indiana', 'Iowa', 'Kansas', 'Kentucky', 'Louisiana', 'Maine', 'Maryland', 'Massachusetts', 'Michigan', 'Minnesota', 'Mississippi', 'Missouri', 'Montana', 'Nebraska', 'Nevada', 'New Hampshire', 'New Jersey', 'New Mexico', 'New York', 'North Carolina', 'North Dakota', 'Ohio', 'Oklahoma', 'Oregon', 'Pennsylvania', 'Rhode Island', 'South Carolina', 'South Dakota', 'Tennessee', 'Texas', 'Utah', 'Vermont', 'Virginia', 'Washington', 'West Virginia', 'Wisconsin', 'Wyoming']
>>> [s.upper() for s in states]
['ALABAMA', 'ALASKA', 'ARIZONA', 'ARKANSAS', 'CALIFORNIA', 'COLORADO', 'CONNECTICUT', 'DELAWARE', 'FLORIDA', 'GEORGIA', 'HAWAII', 'IDAHO', 'ILLINOIS', 'INDIANA', 'IOWA', 'KANSAS', 'KENTUCKY', 'LOUISIANA', 'MAINE', 'MARYLAND', 'MASSACHUSETTS', 'MICHIGAN', 'MINNESOTA', 'MISSISSIPPI', 'MISSOURI', 'MONTANA', 'NEBRASKA', 'NEVADA', 'NEW HAMPSHIRE', 'NEW JERSEY', 'NEW MEXICO', 'NEWYORK', 'NORTH CAROLINA', 'NORTH DAKOTA', 'OHIO', 'OKLAHOMA', 'OREGON', 'PENNSYLVANIA', 'RHODE ISLAND', 'SOUTH CAROLINA', 'SOUTH DAKOTA', 'TENNESSEE', 'TEXAS', 'UTAH', 'VERMONT', 'VIRGINIA', 'WASHINGTON', 'WEST VIRGINIA', 'WISCONSIN', 'WYOMING']
>>> [s.replace(' ', '-') for s in states if ' ' in s]
['New-Hampshire', 'New-Jersey', 'New-Mexico', 'New-York', 'North-Carolina', 'North-Dakota', 'Rhode-Island', 'South-Carolina', 'South-Dakota', 'West-Virginia']
>>> [s for s in states if 'a' not in s.lower()]
['Connecticut', 'Illinois', 'Kentucky', 'Mississippi', 'Missouri', 'New Jersey', 'New Mexico', 'New York', 'Ohio', 'Oregon', 'Tennessee', 'Vermont', 'Wisconsin', 'Wyoming']
>>> [len(s) for s in states]
[7, 6, 7, 8, 10, 8, 11, 8, 7, 7, 6, 5, 8, 7, 4, 6, 8, 9, 5, 8, 13, 8, 9, 11, 8, 7, 8, 6, 13, 10, 10, 8, 14, 12, 4, 8, 6, 12, 12, 14, 12, 9, 5, 4, 7, 8, 10, 13, 9, 7]
>>> max([len(s) for s in states])
14
>>> [(len(s), s) for s in states]
[(7, 'Alabama'), (6, 'Alaska'), (7, 'Arizona'), (8, 'Arkansas'), (10, 'California'), (8, 'Colorado'),
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(11, 'Connecticut'), (8, 'Delaware'), (7, 'Florida'), (7, 'Georgia'), (6, 'Hawaii'), (5, 'Idaho'),
(8, 'Illinois'), (7, 'Indiana'), (4, 'Iowa'), (6, 'Kansas'), (8, 'Kentucky'), (9, 'Louisiana'), (5,
'Maine'), (8, 'Maryland'), (13, 'Massachusetts'), (8, 'Michigan'), (9, 'Minnesota'), (11,
'Mississippi'), (8, 'Missouri'), (7, 'Montana'), (8, 'Nebraska'), (6, 'Nevada'), (13, 'New
Hampshire'), (10, 'New Jersey'), (10, 'New Mexico'), (8, 'New York'), (14, 'North Carolina'), (12,
'North Dakota'), (4, 'Ohio'), (8, 'Oklahoma'), (6, 'Oregon'), (12, 'Pennsylvania'), (12, 'Rhode
Island'), (14, 'South Carolina'), (12, 'South Dakota'), (9, 'Tennessee'), (5, 'Texas'), (4, 'Utah'),
(7, 'Vermont'), (8, 'Virginia'), (10, 'Washington'), (13, 'West Virginia'), (9, 'Wisconsin'), (7,
'Wyoming')]
>>> sorted([(len(s), s) for s in states])
[(4, 'Iowa'), (4, 'Ohio'), (4, 'Utah'), (5, 'Idaho'), (5, 'Maine'), (5, 'Texas'), (6, 'Alaska'), (6,
'Hawaii'), (6, 'Kansas'), (6, 'Nevada'), (6, 'Oregon'), (7, 'Alabama'), (7, 'Arizona'), (7,
'Florida'), (7, 'Georgia'), (7, 'Indiana'), (7, 'Montana'), (7, 'Vermont'), (7, 'Wyoming'), (8,
'Arkansas'), (8, 'Colorado'), (8, 'Delaware'), (8, 'Illinois'), (8, 'Kentucky'), (8, 'Maryland'), (8,
'Michigan'), (8, 'Missouri'), (8, 'Nebraska'), (8, 'New York'), (8, 'Oklahoma'), (8, 'Virginia'), (9,
'Louisiana'), (9, 'Minnesota'), (9, 'Tennessee'), (9, 'Wisconsin'), (10, 'California'), (10, 'New
Jersey'), (10, 'New Mexico'), (10, 'Washington'), (11, 'Connecticut'), (11, 'Mississippi'), (12,
'North Dakota'), (12, 'Pennsylvania'), (12, 'Rhode Island'), (12, 'South Dakota'), (13,
'Massachusetts'), (13, 'New Hampshire'), (13, 'West Virginia'), (14, 'North Carolina'), (14, 'South
Carolina')]
>>> sorted([2, 5, 22, 8, 145, 9])
[2, 5, 8, 9, 22, 145]
>>> sorted([2, 5, 22, 8, 145, 9], reverse=True)
[145, 22, 9, 8, 5, 2]
>>> sorted([(len(s), s) for s in states], reverse=True)
[(14, 'South Carolina'), (14, 'North Carolina'), (13, 'West Virginia'), (13, 'New Hampshire'), (13,
'Massachusetts'), (12, 'South Dakota'), (12, 'Rhode Island'), (12, 'Pennsylvania'), (12, 'North
Dakota'), (11, 'Mississippi'), (11, 'Connecticut'), (10, 'Washington'), (10, 'New Mexico'), (10, 'New
Jersey'), (10, 'California'), (9, 'Wisconsin'), (9, 'Tennessee'), (9, 'Minnesota'), (9, 'Louisiana'),
(8, 'Virginia'), (8, 'Oklahoma'), (8, 'New York'), (8, 'Nebraska'), (8, 'Missouri'), (8, 'Michigan'),
(8, 'Maryland'), (8, 'Kentucky'), (8, 'Illinois'), (8, 'Delaware'), (8, 'Colorado'), (8, 'Arkansas'),
(7, 'Wyoming'), (7, 'Vermont'), (7, 'Montana'), (7, 'Indiana'), (7, 'Georgia'), (7, 'Florida'), (7,
'Arizona'), (7, 'Alabama'), (6, 'Oregon'), (6, 'Nevada'), (6, 'Kansas'), (6, 'Hawaii'), (6,
'Alaska'), (5, 'Texas'), (5, 'Maine'), (5, 'Idaho'), (4, 'Utah'), (4, 'Ohio'), (4, 'Iowa')]
>>> sorted([(len(s), s) for s in states], reverse=True)[:5]
[(14, 'South Carolina'), (14, 'North Carolina'), (13, 'West Virginia'), (13, 'New Hampshire'), (13,
'Massachusetts')]

>>> capital = {'al':'Montgomery', 'ak':'Juneau', 'az':'Phoenix',
...             'ar':'Little Rock', 'ca':'Sacramento', 'co':'Denver',
...             'ct':'Hartford', 'de':'Dover', 'fl':'Tallahassee', 'ga':'Atlanta',
...             'hi':'Honolulu', 'id':'Boise', 'il':'Springfield',
...             'in':'Indianapolis', 'ia':'Des Moines', 'ks':'Topeka',
...             'ky':'Frankfort', 'la':'Baton Rouge', 'me':'Augusta',
...             'md':'Annapolis', 'ma':'Boston', 'mi':'Lansing', 'mn':'St. Paul',
...             'ms':'Jackson', 'mo':'Jefferson City', 'mt':'Helena',
...             'ne':'Lincoln', 'nv':'Carson City', 'nh':'Concord', 'nj':'Trenton',
...             'nm':'Santa Fe', 'ny':'Albany', 'nc':'Raleigh', 'nd':'Bismarck',
...             'oh':'Columbus', 'ok':'Oklahoma City', 'or':'Salem',
...             'pa':'Harrisburg', 'ri':'Providence', 'sc':'Columbia',
...             'sd':'Pierre', 'tn':'Nashville', 'tx':'Austin',
...             'ut':'Salt Lake City', 'vt':'Montpelier', 'va':'Richmond',
...             'wa':'Olympia', 'wv':'Charleston', 'wi':'Madison', 'wy':'Cheyenne'}
>>>
>>>
>>> capital['pa']
'Harrisburg'
>>> capital.values()
dict_values(['Montgomery', 'Juneau', 'Phoenix', 'Little Rock', 'Sacramento', 'Denver', 'Hartford',
'Dover', 'Tallahassee', 'Atlanta', 'Honolulu', 'Boise', 'Springfield', 'Indianapolis', 'Des Moines',
'Topeka', 'Frankfort', 'Baton Rouge', 'Augusta', 'Annapolis', 'Boston', 'Lansing', 'St. Paul',
'Jackson', 'Jefferson City', 'Helena', 'Lincoln', 'Carson City', 'Concord', 'Trenton', 'Santa Fe',
'Albany', 'Raleigh', 'Bismarck', 'Columbus', 'Oklahoma City', 'Salem', 'Harrisburg', 'Providence',

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'Columbia', 'Pierre', 'Nashville', 'Austin', 'Salt Lake City', 'Montpelier', 'Richmond', 'Olympia',
'Charleston', 'Madison', 'Cheyenne'])
>>>
>>> capital.keys()
dict_keys(['al', 'ak', 'az', 'ar', 'ca', 'co', 'ct', 'de', 'fl', 'ga', 'hi', 'id', 'il', 'in', 'ia',
'ks', 'ky', 'la', 'me', 'md', 'ma', 'mi', 'mn', 'ms', 'mo', 'mt', 'ne', 'nv', 'nh', 'nj', 'nm', 'ny',
'nc', 'nd', 'oh', 'ok', 'or', 'pa', 'ri', 'sc', 'sd', 'tn', 'tx', 'ut', 'vt', 'va', 'wa', 'wv', 'wi',
'wy'])
>>> capital.values()
dict_values(['Montgomery', 'Juneau', 'Phoenix', 'Little Rock', 'Sacramento', 'Denver', 'Hartford',
'Dover', 'Tallahassee', 'Atlanta', 'Honolulu', 'Boise', 'Springfield', 'Indianapolis', 'Des Moines',
'Topeka', 'Frankfort', 'Baton Rouge', 'Augusta', 'Annapolis', 'Boston', 'Lansing', 'St. Paul',
'Jackson', 'Jefferson City', 'Helena', 'Lincoln', 'Carson City', 'Concord', 'Trenton', 'Santa Fe',
'Albany', 'Raleigh', 'Bismarck', 'Columbus', 'Oklahoma City', 'Salem', 'Harrisburg', 'Providence',
'Columbia', 'Pierre', 'Nashville', 'Austin', 'Salt Lake City', 'Montpelier', 'Richmond', 'Olympia',
'Charleston', 'Madison', 'Cheyenne'])
>>> sorted(capital.values())
['Albany', 'Annapolis', 'Atlanta', 'Augusta', 'Austin', 'Baton Rouge', 'Bismarck', 'Boise', 'Boston',
'Carson City', 'Charleston', 'Cheyenne', 'Columbia', 'Columbus', 'Concord', 'Denver', 'Des Moines',
'Dover', 'Frankfort', 'Harrisburg', 'Hartford', 'Helena', 'Honolulu', 'Indianapolis', 'Jackson',
'Jefferson City', 'Juneau', 'Lansing', 'Lincoln', 'Little Rock', 'Madison', 'Montgomery',
'Montpelier', 'Nashville', 'Oklahoma City', 'Olympia', 'Phoenix', 'Pierre', 'Providence', 'Raleigh',
'Richmond', 'Sacramento', 'Salem', 'Salt Lake City', 'Santa Fe', 'Springfield', 'St. Paul',
'Tallahassee', 'Topeka', 'Trenton']
>>> sorted(capital.values(), reverse=True)
['Trenton', 'Topeka', 'Tallahassee', 'St. Paul', 'Springfield', 'Santa Fe', 'Salt Lake City',
'Salem', 'Sacramento', 'Richmond', 'Raleigh', 'Providence', 'Pierre', 'Phoenix', 'Olympia', 'Oklahoma
City', 'Nashville', 'Montpelier', 'Montgomery', 'Madison', 'Little Rock', 'Lincoln', 'Lansing',
'Juneau', 'Jefferson City', 'Jackson', 'Indianapolis', 'Honolulu', 'Helena', 'Hartford',
'Harrisburg', 'Frankfort', 'Dover', 'Des Moines', 'Denver', 'Concord', 'Columbus', 'Columbia',
'Cheyenne', 'Charleston', 'Carson City', 'Boston', 'Boise', 'Bismarck', 'Baton Rouge', 'Austin',
'Augusta', 'Atlanta', 'Annapolis', 'Albany']
>>> capital['ma']
'Boston'
>>> capital.items()
dict_items([('al', 'Montgomery'), ('ak', 'Juneau'), ('az', 'Phoenix'), ('ar', 'Little Rock'), ('ca',
'Sacramento'), ('co', 'Denver'), ('ct', 'Hartford'), ('de', 'Dover'), ('fl', 'Tallahassee'), ('ga',
'Atlanta'), ('hi', 'Honolulu'), ('id', 'Boise'), ('il', 'Springfield'), ('in', 'Indianapolis'),
('ia', 'Des Moines'), ('ks', 'Topeka'), ('ky', 'Frankfort'), ('la', 'Baton Rouge'), ('me',
'Augusta'), ('md', 'Annapolis'), ('ma', 'Boston'), ('mi', 'Lansing'), ('mn', 'St. Paul'), ('ms',
'Jackson'), ('mo', 'Jefferson City'), ('mt', 'Helena'), ('ne', 'Lincoln'), ('nv', 'Carson City'),
('nh', 'Concord'), ('nj', 'Trenton'), ('nm', 'Santa Fe'), ('ny', 'Albany'), ('nc', 'Raleigh'), ('nd',
'Bismarck'), ('oh', 'Columbus'), ('ok', 'Oklahoma City'), ('or', 'Salem'), ('pa', 'Harrisburg'),
('ri', 'Providence'), ('sc', 'Columbia'), ('sd', 'Pierre'), ('tn', 'Nashville'), ('tx', 'Austin'),
('ut', 'Salt Lake City'), ('vt', 'Montpelier'), ('va', 'Richmond'), ('wa', 'Olympia'), ('wv',
'Charleston'), ('wi', 'Madison'), ('wy', 'Cheyenne')])
>>> [(s,c) for (s,c) in capital.items() if c=='Concord']
[('nh', 'Concord')]
>>> for x in capital:
...     print(x)
...
al
ak
az
ar
ca
co
ct
# ... results clipped! ...
wv
wi
wy
>>> [x for x in capital if 'k' in x]
['ak', 'ks', 'ky', 'ok']
>>> [x for x in capital.values() if 'k' in x]

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['Little Rock', 'Topeka', 'Frankfort', 'Jackson', 'Bismarck', 'Oklahoma City', 'Salt Lake City']
>>> [x for x in capital.keys() if 'k' in x]
['ak', 'ks', 'ky', 'ok']
>>> [x for x in capital.items()]
[('al', 'Montgomery'), ('ak', 'Juneau'), ('az', 'Phoenix'), ('ar', 'Little Rock'), ('ca',
'Sacramento'), ('co', 'Denver'), ('ct', 'Hartford'), ('de', 'Dover'), ('fl', 'Tallahassee'), ('ga',
'Atlanta'), ('hi', 'Honolulu'), ('id', 'Boise'), ('il', 'Springfield'), ('in', 'Indianapolis'),
('ia', 'Des Moines'), ('ks', 'Topeka'), ('ky', 'Frankfort'), ('la', 'Baton Rouge'), ('me',
'Augusta'), ('md', 'Annapolis'), ('ma', 'Boston'), ('mi', 'Lansing'), ('mn', 'St. Paul'), ('ms',
'Jackson'), ('mo', 'JeffersonCity'), ('mt', 'Helena'), ('ne', 'Lincoln'), ('nv', 'Carson City'),
('nh', 'Concord'), ('nj', 'Trenton'), ('nm', 'Santa Fe'), ('ny', 'Albany'), ('nc', 'Raleigh'), ('nd',
'Bismarck'), ('oh', 'Columbus'), ('ok', 'Oklahoma City'), ('or', 'Salem'), ('pa', 'Harrisburg'),
('ri', 'Providence'), ('sc', 'Columbia'), ('sd', 'Pierre'), ('tn', 'Nashville'), ('tx', 'Austin'),
('ut', 'Salt Lake City'), ('vt', 'Montpelier'), ('va', 'Richmond'), ('wa', 'Olympia'), ('wv',
'Charleston'), ('wi', 'Madison'), ('wy', 'Cheyenne')]
>>> capital.items()
dict_items([('al', 'Montgomery'), ('ak', 'Juneau'), ('az', 'Phoenix'), ('ar', 'Little Rock'), ('ca',
'Sacramento'), ('co', 'Denver'), ('ct', 'Hartford'), ('de', 'Dover'), ('fl', 'Tallahassee'), ('ga',
'Atlanta'), ('hi', 'Honolulu'), ('id', 'Boise'), ('il', 'Springfield'), ('in', 'Indianapolis'),
('ia', 'Des Moines'), ('ks', 'Topeka'), ('ky', 'Frankfort'), ('la', 'Baton Rouge'), ('me',
'Augusta'), ('md', 'Annapolis'), ('ma', 'Boston'), ('mi', 'Lansing'), ('mn', 'St. Paul'), ('ms',
'Jackson'), ('mo', 'Jefferson City'), ('mt', 'Helena'), ('ne', 'Lincoln'), ('nv', 'Carson City'),
('nh', 'Concord'), ('nj', 'Trenton'), ('nm', 'Santa Fe'), ('ny', 'Albany'), ('nc', 'Raleigh'), ('nd',
'Bismarck'), ('oh', 'Columbus'), ('ok', 'Oklahoma City'), ('or', 'Salem'), ('pa', 'Harrisburg'),
('ri', 'Providence'), ('sc', 'Columbia'), ('sd', 'Pierre'), ('tn', 'Nashville'), ('tx', 'Austin'),
('ut', 'Salt Lake City'), ('vt', 'Montpelier'), ('va', 'Richmond'), ('wa', 'Olympia'), ('wv',
'Charleston'), ('wi', 'Madison'), ('wy', 'Cheyenne')])
>>> [(s,c) for (s,c) in capital.items() if 'k' in s and 'k' in c]
[('ks', 'Topeka'), ('ky', 'Frankfort'), ('ok', 'Oklahoma City')]

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>>> popul={'al':4833722, 'ak':735132, 'az':6626624, 'ar':2959373, 'ca':38332521,
...        'co':5268367, 'ct':3596080, 'de':925749, 'fl':19552860, 'ga':9992167,
...        'hi':1404054, 'id':1612136, 'il':12882135, 'in':6570902, 'ia':3090416,
...        'ks':2893957, 'ky':4395295, 'la':4625470, 'me':1328302, 'md':5928814,
...        'ma':6692824, 'mi':9895622, 'mn':5420380, 'ms':2991207, 'mo':6021988,
...        'mt':1015165, 'ne':1868516, 'nv':2790136, 'nh':1323459, 'nj':8899339,
...        'nm':2085287, 'ny':19651127, 'nc':9848060, 'nd':723393, 'oh':11570808,
...        'ok':3850568, 'or':3930065, 'pa':12773801, 'ri':1051511, 'sc':4774839,
...        'sd':844877, 'tn':6495978, 'tx':26448193, 'ut':2900872, 'vt':626630,
...        'va':8260405, 'wa':6971406, 'wv':1854304, 'wi':5742713, 'wy':582658}
>>>
>>> popul['pa']
12773801
>>> for s in popul:
...     if popul[s] > popul['pa']:
...         print(s, popul[s])
...
ca 38332521
fl 19552860
il 12882135
ny 19651127
tx 26448193
>>> [s for s in popul if popul[s] > popul['pa']]
['ca', 'fl', 'il', 'ny', 'tx']
>>> [(s, popul[s]) for s in popul if popul[s] > popul['pa']]
[('ca', 38332521), ('fl', 19552860), ('il', 12882135), ('ny', 19651127), ('tx', 26448193)]
>>> [(s,p) for (s,p) in popul.items() if p > popul['pa']]
[('ca', 38332521), ('fl', 19552860), ('il', 12882135), ('ny', 19651127), ('tx', 26448193)]
>>> [(s,p, p/popul['pa']) for (s,p) in popul.items() if p > popul['pa']]
[('ca', 38332521, 3.0008703752313033), ('fl', 19552860, 1.530700219926708), ('il', 12882135,
1.00848095253715), ('ny', 19651127, 1.5383930750134591), ('tx', 26448193, 2.0705029771483052)]
>>> sorted([(p/popul['pa'], s,p) for (s,p) in popul.items() if p > popul['pa']])

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[(1.00848095253715, 'il', 12882135), (1.530700219926708, 'fl', 19552860), (1.5383930750134591, 'ny', 19651127), (2.0705029771483052, 'tx', 26448193), (3.0008703752313033, 'ca', 38332521)]
>>> sorted([(p,popul['pa'], s,p) for (s,p) in popul.items() if p > popul['pa']], reverse=True)
[(3.0008703752313033, 'ca', 38332521), (2.0705029771483052, 'tx', 26448193), (1.5383930750134591, 'ny', 19651127), (1.530700219926708, 'fl', 19552860), (1.00848095253715, 'il', 12882135)]
>>> [(s,p) for (s,p) in popul.items() if p > popul['pa']]
[('ca', 38332521), ('fl', 19552860), ('il', 12882135), ('ny', 19651127), ('tx', 26448193)]
>>> dict([(s,p) for (s,p) in popul.items() if p > popul['pa']])
{'ca': 38332521, 'fl': 19552860, 'il': 12882135, 'ny': 19651127, 'tx': 26448193}

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>>> footuple = [('a', 'apple'), ('b', 'banana'), ('c', 'carrot')]
>>> footuple
[('a', 'apple'), ('b', 'banana'), ('c', 'carrot')]
>>> [y for (x,y) in footuple]
['a', 'b', 'c']
>>> [(y,x) for (x,y) in footuple]
[('apple', 'a'), ('banana', 'b'), ('carrot', 'c')]
>>> dict(footuple)
{'a': 'apple', 'b': 'banana', 'c': 'carrot'}
>>> foodict = dict(footuple)
>>> foodict
{'a': 'apple', 'b': 'banana', 'c': 'carrot'}
>>> foodict['a']
'apple'

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Copy over gettysburg address string

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>>> getty
'Four score and seven years ago our fathers brought forth on this continent a new nation, conceived
in liberty, and dedicated to the proposition that all men are created equal.\n\nNow we are engaged in
a great civil war, testing whether that nation, or any nation, so conceived and so dedicated, can
long endure. We are met on a great battle-field of that war. We have come to dedicate a portion of
that field, as a final resting place for those who here gave their lives that that nation might live.
It is altogether fitting and proper that we should do this.\n\nBut, in a larger sense, we can not
dedicate, we can not consecrate, we can not hallow this ground. The brave men, living and dead, who
struggled here, have consecrated it, far above our poor power to add or detract. The world will
little note, nor long remember what we say here, but it can never forget what they did here. It is for
us the living, rather, to be dedicated here to the unfinished work which they who fought here have
thus far so nobly advanced. It is rather for us to be here dedicated to the great task remaining
before us - that from these honored dead we take increased devotion to that cause for which they gave
the last full measure of devotion - that we here highly resolve that these dead shall not have died in
vain - that this nation, under God, shall have a new birth of freedom - and that government of the
people, by the people, for the people, shall not perish from the earth.'
>>> getty_toks = getty.split()
>>> getty_toks[:30]
['Four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',
'continent', 'a', 'new', 'nation', 'conceived', 'in', 'liberty', 'and', 'dedicated', 'to', 'the',
'proposition', 'that', 'all', 'men', 'are', 'created', 'equal.']
>>> len(getty_toks)
275
>>> [t for t in getty_toks if len(t) <=3]
['and', 'ago', 'our', 'on', 'a', 'new', 'in', 'and', 'to', 'the', 'all', 'men', 'are', 'Now', 'we',
'are', 'in', 'a', 'or', 'any', 'so', 'and', 'so', 'can', 'We', 'are', 'met', 'on', 'a', 'of', 'We',
'to', 'a', 'of', 'as', 'a', 'for', 'who', 'It', 'is', 'and', 'we', 'do', 'in', 'a', 'we', 'can',
'not', 'we', 'can', 'not', 'we', 'can', 'not', 'The', 'and', 'who', 'it', 'far', 'our', 'to', 'add',
'or', 'The', 'nor', 'we', 'say', 'but', 'it', 'can', 'did', 'It', 'is', 'for', 'us', 'the', 'to',
'be', 'to', 'the', 'who', 'far', 'so', 'It', 'is', 'for', 'us', 'to', 'be', 'to', 'the', 'us', '-',
'we', 'to', 'for', 'the', 'of', '-', 'we', 'not', 'in', '-', 'a', 'new', 'of', '-', 'and', 'of',
'the', 'by', 'the', 'for', 'the', 'not', 'the']
>>> [t for t in getty_toks if len(t) > 5]
['fathers', 'brought', 'continent', 'nation', 'conceived', 'liberty', 'dedicated', 'proposition',
'created', 'equal.', 'engaged', 'testing', 'whether', 'nation', 'nation', 'conceived',
'dedicated', 'endure.', 'battle-field', 'dedicate', 'portion', 'field', 'resting', 'nation',
'altogether', 'fitting', 'proper', 'should', 'larger', 'sense', 'dedicate', 'consecrate',

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'hallow', 'ground.', 'living', 'struggled', 'consecrated', 'detract.', 'little', 'remember',
'forget', 'living,', 'rather,', 'dedicated', 'unfinished', 'fought', 'advanced.', 'rather',
'dedicated', 'remaining', 'before', 'honored', 'increased', 'devotion', 'measure', 'devotion',
'highly', 'resolve', 'nation,', 'freedom', 'government', 'people,', 'people,', 'people,', 'perish',
'earth.']]
>>> [(len(t), t) for t in getty_toks]
[(4, 'Four'), (5, 'score'), (3, 'and'), (5, 'seven'), (5, 'years'), (3, 'ago'), (3, 'our'), (7,
'fathers'), (7, 'brought'), (5, 'forth'), (2, 'on'), (4, 'this'), (9, 'continent'), (1, 'a'), (3,
'new'), (7, 'nation,'), (9, 'conceived'), (2, 'in'), (8, 'liberty,'), (3, 'and'), (9, 'dedicated'),
(2, 'to'), (3, 'the'), (11, 'proposition'), (4, 'that'), (3, 'all'), (3, 'men'), (3, 'are'), (7,
'created'), (6, 'equal.'), (3, 'Now'), (2, 'we'), (3, 'are'), (7, 'engaged'), (2, 'in'), (1, 'a'),
(5, 'great'), (5, 'civil'), (4, 'war,'), (7, 'testing'), (7, 'whether'), (4, 'that'), (7, 'nation,'),
(2, 'or'), (3, 'any'), (7, 'nation,'), (2, 'so'), (9, 'conceived'), (3, 'and'), (2, 'so'), (10,
'dedicated,'), (3, 'can'), (4, 'long'), (7, 'endure.'), (2, 'We'), (3, 'are'), (3, 'met'), (2, 'on'),
(1, 'a'), (5, 'great'), (12, 'battle-field'), (2, 'of'), (4, 'that'), (4, 'war.'), (2, 'We'), (4,
'have'), (4, 'come'), (2, 'to'), (8, 'dedicate'), (1, 'a'), (7, 'portion'), (2, 'of'), (4, 'that'),
(6, 'field,'), (2, 'as'), (1, 'a'), (5, 'final'), (7, 'resting'), (5, 'place'), (3, 'for'), (5,
'those'), (3, 'who'), (4, 'here'), (4, 'gave'), (5, 'their'), (5, 'lives'), (4, 'that'), (4, 'that'),
(6, 'nation'), (5, 'might'), (5, 'live.'), (2, 'It'), (2, 'is'), (10, 'altogether'), (7, 'fitting'),
(3, 'and'), (6, 'proper'), (4, 'that'), (2, 'we'), (6, 'should'), (2, 'do'), (5, 'this.'), (4,
'But,'), (2, 'in'), (1, 'a'), (6, 'larger'), (6, 'sense,'), (2, 'we'), (3, 'can'), (3, 'not'), (9,
'dedicate,'), (2, 'we'), (3, 'can'), (3, 'not'), (11, 'consecrate,'), (2, 'we'), (3, 'can'), (3,
'not'), (6, 'hallow'), (4, 'this'), (7, 'ground.'), (3, 'The'), (5, 'brave'), (4, 'men,'), (6,
'living'), (3, 'and'), (5, 'dead,'), (3, 'who'), (9, 'struggled'), (5, 'here,'), (4, 'have'), (11,
'consecrated'), (3, 'it,'), (3, 'far'), (5, 'above'), (3, 'our'), (4, 'poor'), (5, 'power'), (2,
'to'), (3, 'add'), (2, 'or'), (8, 'detract.'), (3, 'The'), (5, 'world'), (4, 'will'), (6, 'little'),
(5, 'note,'), (3, 'nor'), (4, 'long'), (8, 'remember'), (4, 'what'), (2, 'we'), (3, 'say'), (5,
'here,'), (3, 'but'), (2, 'it'), (3, 'can'), (5, 'never'), (6, 'forget'), (4, 'what'), (4, 'they'),
(3, 'did'), (5, 'here.'), (2, 'It'), (2, 'is'), (3, 'for'), (2, 'us'), (3, 'the'), (7, 'living,'),
(7, 'rather,'), (2, 'to'), (2, 'be'), (9, 'dedicated'), (4, 'here'), (2, 'to'), (3, 'the'), (10,
'unfinished'), (4, 'work'), (5, 'which'), (4, 'they'), (3, 'who'), (6, 'fought'), (4, 'here'), (4,
'have'), (4, 'thus'), (3, 'far'), (2, 'so'), (5, 'nobly'), (9, 'advanced.'), (2, 'It'), (2, 'is'), (6,
'rather'), (3, 'for'), (2, 'us'), (2, 'to'), (2, 'be'), (4, 'here'), (9, 'dedicated'), (2, 'to'), (3,
'the'), (5, 'great'), (4, 'task'), (9, 'remaining'), (6, 'before'), (2, 'us'), (1, '-'), (4, 'that'),
(4, 'from'), (5, 'these'), (7, 'honored'), (4, 'dead'), (2, 'we'), (4, 'take'), (9, 'increased'), (8,
'devotion'), (2, 'to'), (4, 'that'), (5, 'cause'), (3, 'for'), (5, 'which'), (4, 'they'), (4, 'gave'),
(3, 'the'), (4, 'last'), (4, 'full'), (7, 'measure'), (2, 'of'), (8, 'devotion'), (1, '-'), (4,
'that'), (2, 'we'), (4, 'here'), (6, 'highly'), (7, 'resolve'), (4, 'that'), (5, 'these'), (4,
'dead'), (5, 'shall'), (3, 'not'), (4, 'have'), (4, 'died'), (2, 'in'), (4, 'vain'), (1, '-'), (4,
'that'), (4, 'this'), (7, 'nation,'), (5, 'under'), (4, 'God,'), (5, 'shall'), (4, 'have'), (1, 'a'),
(3, 'new'), (5, 'birth'), (2, 'of'), (7, 'freedom'), (1, '-'), (3, 'and'), (4, 'that'), (10,
'government'), (2, 'of'), (3, 'the'), (7, 'people,'), (2, 'by'), (3, 'the'), (7, 'people,'), (3,
'for'), (3, 'the'), (7, 'people,'), (5, 'shall'), (3, 'not'), (6, 'perish'), (4, 'from'), (3, 'the'),
(6, 'earth.']]
>>> sorted([(len(t), t) for t in getty_toks], reverse=True)[:10]
[(12, 'battle-field'), (11, 'proposition'), (11, 'consecrated'), (11, 'consecrate,'), (10,
'unfinished'), (10, 'government'), (10, 'dedicated,'), (10, 'altogether'), (9, 'struggled'), (9,
'remaining')]
>>> [t for t in getty_toks if not t.isalnum()]
['nation,', 'liberty,', 'equal.', 'war,', 'nation,', 'nation,', 'dedicated,', 'endure.', 'battle-
field', 'war.', 'field,', 'live.', 'this.', 'But,', 'sense,', 'dedicate,', 'consecrate,', 'ground.',
'men,', 'dead,', 'here,', 'it,', 'detract.', 'note,', 'here,', 'here.', 'living,', 'rather,',
'advanced.', '-', '-', '-', 'nation,', 'God,', '-', 'people,', 'people,', 'people,', 'earth.']]
>>> [t for t in getty_toks if t[0].isupper()]
['Four', 'Now', 'We', 'We', 'It', 'But,', 'The', 'The', 'It', 'It', 'God,']
>>> [t for t in getty_toks if len(t) > 10 and '-' not in t]
['proposition', 'consecrate,', 'consecrated']
>>> [t for t in getty_toks if len(t) > 10 and '-' in t]
['battle-field']
>>> sent = 'We love school.'
>>> sent
'We love school.'
>>> sent.split(' ')
['We', 'love', 'school.']]
>>> sent.split()

```



```

['We', 'love', 'school.']
>>> sent.replace('.', '')
'We love school'
>>> sent.replace('.', '').split()
['We', 'love', 'school']
>>> sent = 'We love school, and we love our teacher.'
>>> sent.replace('.', '').split()
['We', 'love', 'school,', 'and', 'we', 'love', 'our', 'teacher']
>>> sent.replace('.', '').replace(',','').split()
['We', 'love', 'school', 'and', 'we', 'love', 'our', 'teacher']
>>> getty_toks
['Four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',
'continent', 'a', 'new', 'nation,', 'conceived', 'in', 'liberty,', 'and', 'dedicated', 'to', 'the',
'proposition', 'that', 'all', 'men', 'are', 'created', 'equal.', 'Now', 'we', 'are', 'engaged', 'in',
'a', 'great', 'civil', 'war,', 'testing', 'whether', 'that', 'nation,', 'or', 'any', 'nation,', 'so',
'conceived', 'and', 'so', 'dedicated,', 'can', 'long', 'endure.', 'We', 'are', 'met', 'on', 'a',
'great', 'battle-field', 'of', 'that', 'war.', 'We', 'have', 'come', 'to', 'dedicate', 'a',
'portion', 'of', 'that', 'field,', 'as', 'a', 'final', 'resting', 'place', 'for', 'those', 'who',
'here', 'gave', 'their', 'lives', 'that', 'that', 'nation', 'might', 'live.', 'It', 'is',
'altogether', 'fitting', 'and', 'proper', 'that', 'we', 'should', 'do', 'this.', 'But', 'in', 'a',
'larger', 'sense,', 'we', 'can', 'not', 'dedicate', 'we', 'can', 'not', 'consecrate,', 'we', 'can',
'not', 'hallow', 'this', 'ground.', 'The', 'brave', 'men,', 'living', 'and', 'dead,', 'who',
'struggled', 'here,', 'have', 'consecrated', 'it', 'far', 'above', 'our', 'poor', 'power', 'to',
'add', 'or', 'detract.', 'The', 'world', 'will', 'little', 'note,', 'nor', 'long', 'remember',
'what', 'we', 'say', 'here,', 'but', 'it', 'can', 'never', 'forget', 'what', 'they', 'did', 'here.',
'It', 'is', 'for', 'us', 'the', 'living,', 'rather', 'to', 'be', 'dedicated', 'here', 'to', 'the',
'unfinished', 'work', 'which', 'they', 'who', 'fought', 'here', 'have', 'thus', 'far', 'so', 'nobly',
'advanced.', 'It', 'is', 'rather', 'for', 'us', 'to', 'be', 'here', 'dedicated', 'to', 'the',
'great', 'task', 'remaining', 'before', 'us', '-', 'that', 'from', 'these', 'honored', 'dead', 'we',
'take', 'increased', 'devotion', 'to', 'that', 'cause', 'for', 'which', 'they', 'gave', 'the',
'last', 'full', 'measure', 'of', 'devotion', '-', 'that', 'we', 'here', 'highly', 'resolve', 'that',
'these', 'dead', 'shall', 'not', 'have', 'died', 'in', 'vain', '-', 'that', 'this', 'nation',
'under', 'God,', 'shall', 'have', 'a', 'new', 'birth', 'of', 'freedom', '-', 'and', 'that',
'government', 'of', 'the', 'people,', 'by', 'the', 'people,', 'for', 'the', 'people,', 'shall',
'not', 'perish', 'from', 'the', 'earth.']
>>> getty_toks.replace('.', '')      # can't use .replace() on a list
Traceback (most recent call last):
  File "", line 1, in
    getty_toks.replace('.', '')
    ^^^^^^^^^^^^^^^^^^^^^
AttributeError: 'list' object has no attribute 'replace'
>>> [t.upper() for t in getty_toks]  # use list comprehension
['FOUR', 'SCORE', 'AND', 'SEVEN', 'YEARS', 'AGO', 'OUR', 'FATHERS', 'BROUGHT', 'FORTH', 'ON', 'THIS',
'CONTINENT', 'A', 'NEW', 'NATION,', 'CONCEIVED', 'IN', 'LIBERTY,', 'AND', 'DEDICATED', 'TO', 'THE',
'PROPOSITION', 'THAT', 'ALL', 'MEN', 'ARE', 'CREATED', 'EQUAL.', 'NOW', 'WE', 'ARE', 'ENGAGED', 'IN',
'A', 'GREAT', 'CIVIL', 'WAR,', 'TESTING', 'WHETHER', 'THAT', 'NATION,', 'OR', 'ANY', 'NATION,', 'SO',
'CONCEIVED', 'AND', 'SO', 'DEDICATED,', 'CAN', 'LONG', 'ENDURE.', 'WE', 'ARE', 'MET', 'ON', 'A',
'GREAT', 'BATTLE-FIELD', 'OF', 'THAT', 'WAR.', 'WE', 'HAVE', 'COME', 'TO', 'DEDICATE', 'A',
'PORTION', 'OF', 'THAT', 'FIELD,', 'AS', 'A', 'FINAL', 'RESTING', 'PLACE', 'FOR', 'THOSE', 'WHO',
'HERE', 'GAVE', 'THEIR', 'LIVES', 'THAT', 'THAT', 'NATION', 'MIGHT', 'LIVE.', 'IT', 'IS',
'ALTOGETHER', 'FITTING', 'AND', 'PROPER', 'THAT', 'WE', 'SHOULD', 'DO', 'THIS.', 'BUT', 'IN', 'A',
'LARGER', 'SENSE,', 'WE', 'CAN', 'NOT', 'DEDICATE', 'WE', 'CAN', 'NOT', 'CONSECRATE,', 'WE', 'CAN',
'NOT', 'HALLOW', 'THIS', 'GROUND.', 'THE', 'BRAVE', 'MEN,', 'LIVING', 'AND', 'DEAD,', 'WHO',
'STRUGGLED', 'HERE,', 'HAVE', 'CONSECRATED', 'IT', 'FAR', 'ABOVE', 'OUR', 'POOR', 'POWER', 'TO',
'ADD', 'OR', 'DETRACT.', 'THE', 'WORLD', 'WILL', 'LITTLE', 'NOTE,', 'NOR', 'LONG', 'REMEMBER',
'WHAT', 'WE', 'SAY', 'HERE,', 'BUT', 'IT', 'CAN', 'NEVER', 'FORGET', 'WHAT', 'THEY', 'DID', 'HERE.',
'IT', 'IS', 'FOR', 'US', 'THE', 'LIVING,', 'RATHER', 'TO', 'BE', 'DEDICATED', 'HERE', 'TO', 'THE',
'UNFINISHED', 'WORK', 'WHICH', 'THEY', 'WHO', 'FOUGHT', 'HERE', 'HAVE', 'THUS', 'FAR', 'SO', 'NOBLY',
'ADVANCED.', 'IT', 'IS', 'RATHER', 'FOR', 'US', 'TO', 'BE', 'HERE', 'DEDICATED', 'TO', 'THE',
'GREAT', 'TASK', 'REMAINING', 'BEFORE', 'US', '-', 'THAT', 'FROM', 'THESE', 'HONORED', 'DEAD', 'WE',
'TAKE', 'INCREASED', 'DEVOTION', 'TO', 'THAT', 'CAUSE', 'FOR', 'WHICH', 'THEY', 'GAVE', 'THE',
'LAST', 'FULL', 'MEASURE', 'OF', 'DEVOTION', '-', 'THAT', 'WE', 'HERE', 'HIGHLY', 'RESOLVE', 'THAT',
'THESE', 'DEAD', 'SHALL', 'NOT', 'HAVE', 'DIED', 'IN', 'VAIN', '-', 'THAT', 'THIS', 'NATION',
'UNDER', 'GOD,', 'SHALL', 'HAVE', 'A', 'NEW', 'BIRTH', 'OF', 'FREEDOM', '-', 'AND', 'THAT',
'GOVERNMENT', 'OF', 'THE', 'PEOPLE,', 'BY', 'THE', 'PEOPLE,', 'FOR', 'THE', 'PEOPLE,', 'SHALL',

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'NOT', 'PERISH', 'FROM', 'THE', 'EARTH.']
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>>> [t.replace(',', '') for t in getty_toks]
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['Four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',  
'continent', 'a', 'new', 'nation', 'conceived', 'in', 'liberty', 'and', 'dedicated', 'to', 'the',  
'proposition', 'that', 'all', 'men', 'are', 'created', 'equal.', 'Now', 'we', 'are', 'engaged', 'in',  
'a', 'great', 'civil', 'war', 'testing', 'whether', 'that', 'nation', 'or', 'any', 'nation', 'so',  
'conceived', 'and', 'so', 'dedicated', 'can', 'long', 'endure.', 'We', 'are', 'met', 'on', 'a',  
'great', 'battle-field', 'of', 'that', 'war.', 'We', 'have', 'come', 'to', 'dedicate', 'a',  
'portion', 'of', 'that', 'field', 'as', 'a', 'final', 'resting', 'place', 'for', 'those', 'who',  
'here', 'gave', 'their', 'lives', 'that', 'that', 'nation', 'might', 'live.', 'It', 'is',  
'altogether', 'fitting', 'and', 'proper', 'that', 'we', 'should', 'do', 'this.', 'But', 'in', 'a',  
'larger', 'sense', 'we', 'can', 'not', 'dedicate', 'we', 'can', 'not', 'consecrate', 'we', 'can',  
'not', 'hallow', 'this', 'ground.', 'The', 'brave', 'men', 'living', 'and', 'dead', 'who',  
'struggled', 'here', 'have', 'consecrated', 'it', 'far', 'above', 'our', 'poor', 'power', 'to',  
'add', 'or', 'detract.', 'The', 'world', 'will', 'little', 'note', 'nor', 'long', 'remember', 'what',  
'we', 'say', 'here', 'but', 'it', 'can', 'never', 'forget', 'what', 'they', 'did', 'here.', 'It',  
'is', 'for', 'us', 'the', 'living', 'rather', 'to', 'be', 'dedicated', 'here', 'to', 'the',  
'unfinished', 'work', 'which', 'they', 'who', 'fought', 'here', 'have', 'thus', 'far', 'so', 'nobly',  
'advanced.', 'It', 'is', 'rather', 'for', 'us', 'to', 'be', 'here', 'dedicated', 'to', 'the',  
'great', 'task', 'remaining', 'before', 'us', '-', 'that', 'from', 'these', 'honored', 'dead', 'we',  
'take', 'increased', 'devotion', 'to', 'that', 'cause', 'for', 'which', 'they', 'gave', 'the',  
'last', 'full', 'measure', 'of', 'devotion', '-', 'that', 'we', 'here', 'highly', 'resolve', 'that',  
'these', 'dead', 'shall', 'not', 'have', 'died', 'in', 'vain', '-', 'that', 'this', 'nation',  
'under', 'God', 'shall', 'have', 'a', 'new', 'birth', 'of', 'freedom', '-', 'and', 'that',  
'government', 'of', 'the', 'people', 'by', 'the', 'people', 'for', 'the', 'people', 'shall', 'not',  
'perish', 'from', 'the', 'earth.']
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```
>>> [t.replace(',', '').replace('.', '') for t in getty_toks]
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['Four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',  
'continent', 'a', 'new', 'nation', 'conceived', 'in', 'liberty', 'and', 'dedicated', 'to', 'the',  
'proposition', 'that', 'all', 'men', 'are', 'created', 'equal', 'Now', 'we', 'are', 'engaged', 'in',  
'a', 'great', 'civil', 'war', 'testing', 'whether', 'that', 'nation', 'or', 'any', 'nation', 'so',  
'conceived', 'and', 'so', 'dedicated', 'can', 'long', 'endure', 'We', 'are', 'met', 'on', 'a',  
'great', 'battle-field', 'of', 'that', 'war', 'We', 'have', 'come', 'to', 'dedicate', 'a', 'portion',  
'of', 'that', 'field', 'as', 'a', 'final', 'resting', 'place', 'for', 'those', 'who', 'here', 'gave',  
'their', 'lives', 'that', 'that', 'nation', 'might', 'live', 'It', 'is', 'altogether', 'fitting',  
'and', 'proper', 'that', 'we', 'should', 'do', 'this', 'But', 'in', 'a', 'larger', 'sense', 'we',  
'can', 'not', 'dedicate', 'we', 'can', 'not', 'consecrate', 'we', 'can', 'not', 'hallow', 'this',  
'ground', 'The', 'brave', 'men', 'living', 'and', 'dead', 'who', 'struggled', 'here', 'have',  
'consecrated', 'it', 'far', 'above', 'our', 'poor', 'power', 'to', 'add', 'or', 'detract', 'The',  
'world', 'will', 'little', 'note', 'nor', 'long', 'remember', 'what', 'we', 'say', 'here', 'but',  
'it', 'can', 'never', 'forget', 'what', 'they', 'did', 'here', 'It', 'is', 'for', 'us', 'the',  
'living', 'rather', 'to', 'be', 'dedicated', 'here', 'to', 'the', 'unfinished', 'work', 'which',  
'they', 'who', 'fought', 'here', 'have', 'thus', 'far', 'so', 'nobly', 'advanced', 'It', 'is',  
'rather', 'for', 'us', 'to', 'be', 'here', 'dedicated', 'to', 'the', 'great', 'task', 'remaining',  
'before', 'us', '-', 'that', 'from', 'these', 'honored', 'dead', 'we', 'take', 'increased',  
'devotion', 'to', 'that', 'cause', 'for', 'which', 'they', 'gave', 'the', 'last', 'full', 'measure',  
'of', 'devotion', '-', 'that', 'we', 'here', 'highly', 'resolve', 'that', 'these', 'dead', 'shall',  
'not', 'have', 'died', 'in', 'vain', '-', 'that', 'this', 'nation', 'under', 'God', 'shall', 'have',  
'a', 'new', 'birth', 'of', 'freedom', '-', 'and', 'that', 'government', 'of', 'the', 'people', 'by',  
'the', 'people', 'for', 'the', 'people', 'shall', 'not', 'perish', 'from', 'the', 'earth']
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```
>>> [t.replace(',', '').replace('.', '').lower() for t in getty_toks]
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['four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',  
'continent', 'a', 'new', 'nation', 'conceived', 'in', 'liberty', 'and', 'dedicated', 'to', 'the',  
'proposition', 'that', 'all', 'men', 'are', 'created', 'equal', 'now', 'we', 'are', 'engaged', 'in',  
'a', 'great', 'civil', 'war', 'testing', 'whether', 'that', 'nation', 'or', 'any', 'nation', 'so',  
'conceived', 'and', 'so', 'dedicated', 'can', 'long', 'endure', 'we', 'are', 'met', 'on', 'a',  
'great', 'battle-field', 'of', 'that', 'war', 'we', 'have', 'come', 'to', 'dedicate', 'a', 'portion',  
'of', 'that', 'field', 'as', 'a', 'final', 'resting', 'place', 'for', 'those', 'who', 'here', 'gave',  
'their', 'lives', 'that', 'that', 'nation', 'might', 'live', 'it', 'is', 'altogether', 'fitting',  
'and', 'proper', 'that', 'we', 'should', 'do', 'this', 'but', 'in', 'a', 'larger', 'sense', 'we',  
'can', 'not', 'dedicate', 'we', 'can', 'not', 'consecrate', 'we', 'can', 'not', 'hallow', 'this',  
'ground', 'the', 'brave', 'men', 'living', 'and', 'dead', 'who', 'struggled', 'here', 'have',  
'consecrated', 'it', 'far', 'above', 'our', 'poor', 'power', 'to', 'add', 'or', 'detract', 'the',  
'world', 'will', 'little', 'note', 'nor', 'long', 'remember', 'what', 'we', 'say', 'here', 'but',  
'it', 'can', 'never', 'forget', 'what', 'they', 'did', 'here', 'it', 'is', 'for', 'us', 'the',
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'living', 'rather', 'to', 'be', 'dedicated', 'here', 'to', 'the', 'unfinished', 'work', 'which',
'they', 'who', 'fought', 'here', 'have', 'thus', 'far', 'so', 'nobly', 'advanced', 'it', 'is',
'rather', 'for', 'us', 'to', 'be', 'here', 'dedicated', 'to', 'the', 'great', 'task', 'remaining',
'before', 'us', '-', 'that', 'from', 'these', 'honored', 'dead', 'we', 'take', 'increased',
'devotion', 'to', 'that', 'cause', 'for', 'which', 'they', 'gave', 'the', 'last', 'full', 'measure',
'of', 'devotion', '-', 'that', 'we', 'here', 'highly', 'resolve', 'that', 'these', 'dead', 'shall',
'not', 'have', 'died', 'in', 'vain', '-', 'that', 'this', 'nation', 'under', 'god', 'shall', 'have',
'a', 'new', 'birth', 'of', 'freedom', '-', 'and', 'that', 'government', 'of', 'the', 'people', 'by',
'the', 'people', 'for', 'the', 'people', 'shall', 'not', 'perish', 'from', 'the', 'earth']

>>>